CATERPILLAR®

BEST PRACTICE SERIES

Using VIMS TKPH Feature to Control Heat-Related Tire Failures

Application Maintenance Site Management Component Rebuild Safety MARC Management

CAT®

Global Mining

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1.0 Introduction

With the recent tire shortage many customers are using VIMS TKPH feature to help reduce the number of heat related tire failures. TKPH stands for “Ton-Kilometer Per Hour”, and is a measure used by tire manufacturers to “rate” the ability of their tires to carry a load over a period of time. It is calculated, quite simply, by multiplying the average tire load by the average speed per hour. If the TKPH value for a particular tire is exceeded, the tire overheats and may eventually fail.

The TKPH Monitor allows a site to more carefully manage their tire expense by reducing the number of tires scrapped due to heat separation. The TKPH Monitor constantly monitors tire loading, and provides warnings to the operator and site management, to change the machine’s operation, thus allowing the tires to cool down.

2.0 Best Practice Description

The TKPH Monitor is based on the SAE standard for calculating TKPH. The calculation uses real time data from VIMS including payload, distance traveled, ambient air temperature, speed of the truck, and truck weight. VIMS stores TKPH values in a rolling database with a time base that can be adjusted from one to twelve hours. Most tire manufactures recommend using a four-hour time base. TKPH also makes adjustments when the machine is traveling loaded or unloaded. While unloaded, the front and rear axles to bear equal amounts of machine weight. During this condition, the front tires will typically have the higher TKPH values because the truck empty weight distribution transfers more weight to the front two tires and the rear four tires share the same load. When loaded, however, only one-third of the weight will be on the front axles. This condition will result in the front and rear TKPH values being aligned closer together.

When VIMS calculates that the TKPH limit has been exceeded, the operator will receive an event notification and the truck will become speed limited. The operator will maintain full control, except the machine will not accelerate to maximum speed. Site personnel can customize the speed limit.

While operating the truck below the TKPH Limiting Speed, the operator will have full control of the machine speed up to the Limiting Speed value. As the tires cool off the TKPH Feature will start giving control back to the operator. Once the TKPH values fall below the TKPH deactivation value, the operator will be prompted to turn speed limiting off by pressing the OK key on the keypad. This was done as a safety feature to keep the operator in control of the machine at all times.

How the events for VIMS TKPH work:

1. When the TKPH values remain above the activation threshold for one minute, a "TKPH R WARNING HI" message will be displayed on the message center:

   a. At this point, it is recommended for the operator to reduce drive speed to allow the tires to begin cooling down.

2. Once the truck speed has been reduced below the target speed limit, the "Truck Speed Limited" message will be displayed on the VIMS Message Center:

   a. This is a reminder to the operator that the Machine is speed limited. This message can be "snoozed" to the operator for ten minutes.

   b. When this event is active, the operator will have full functionality of the truck, but can only achieve top speed that has been previously determined by the site.

3. Once the tires have cooled down and VIMS has calculated that the TKPH event is getting ready to be removed, the operator is alerted with a "Speed Limit Off in One Minute" event.

4. This last “Press [OK] to Stop Speed Limiting” message will remain displayed for thirty seconds.

    a. The TKPH Speed Limiting will be immediately deactivated once the [OK] key is pressed.
    b. If the [OK] key is not pressed within thirty seconds the message will reappear at one-minute intervals and be displayed for thirty seconds.
    c. Once the operator pushes the [OK] button, they will have total control of machine speed.

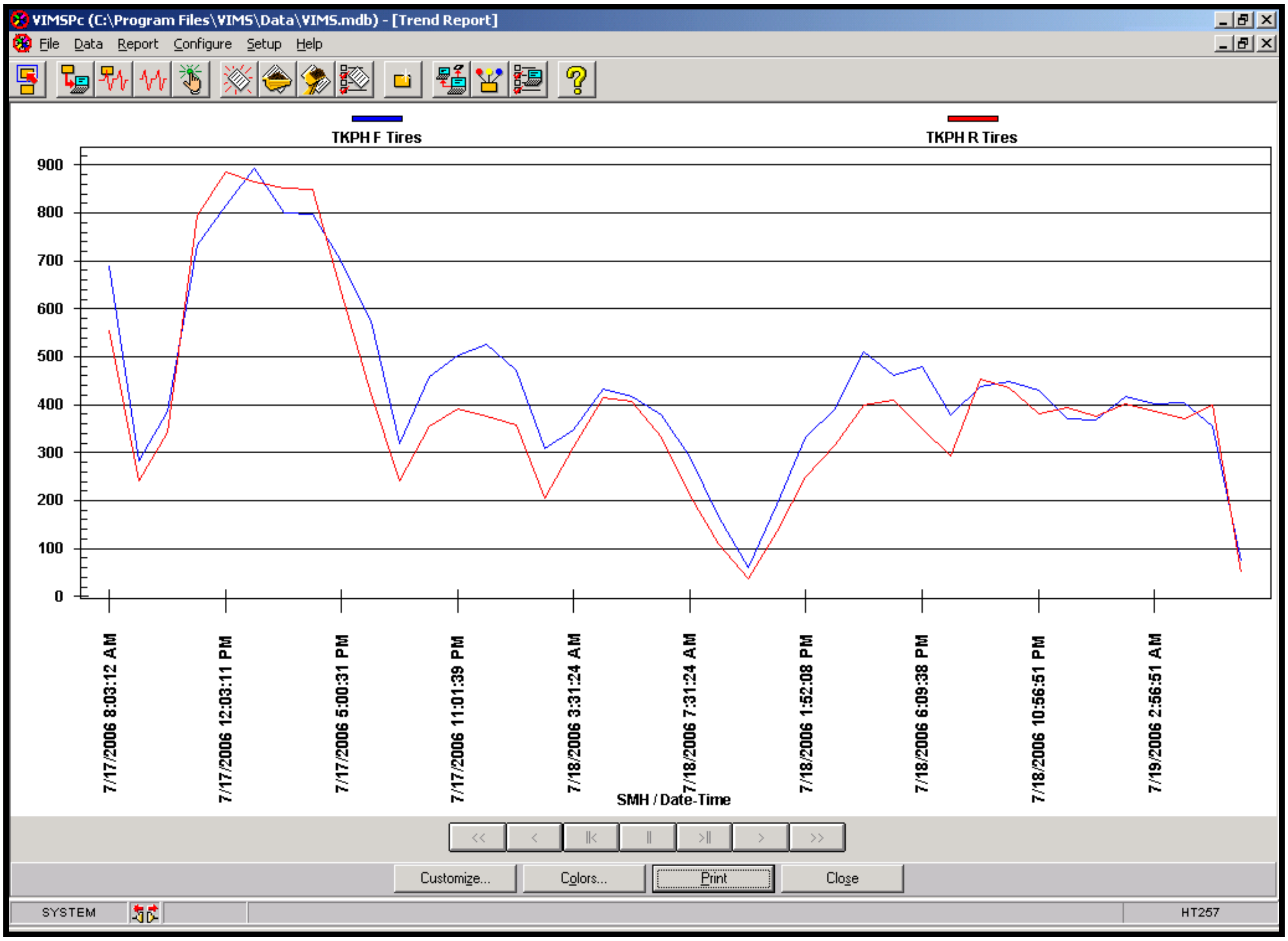
3.0 Implementation Steps

Set up a plan of action to implement VIMS TKPH on your site.

1. Work with the site’s local tire manufacture to determine appropriate settings for the application.
    a. Determine TKPH ratings for the tires presently running at site.
    b. Determine desired speed limit to cool down tires before overheating
    c. Have the tire manufacture run a heat study at your site.
    d. Check tire pressures on a scheduled basis.

2. Work with personnel from Operations and Production in developing a process to use VIMS TKPH on your site.
    a. Set up instructions for operators to follow once they get a TKPH Event.
    b. Work with Production to determine what needs to be done when a truck gets a TKPH event. If the truck is on a long haul cycle, possibly move to a shorter haul cycle until such time that the tires cool down.
    c. Run a daily report on TKPH, by truck, to monitor TKPH values.

Example of a VIMS TKPH daily report



3. In conjunction with the site's tire supplier, a tire study should be performed to determine the appropriate activation and deactivation points to trigger the speed limiting function. The site's tire supplier is the expert in this area and must be consulted.
    a. The site should decide on an appropriate speed limit. The slower the speed, the faster the tires will cool down, but a balance must be met between tire maintenance needs and production needs.
    b. Once satisfied with these limits, the Caterpillar dealer representative, or site's service technician, can set the activation and deactivation points through the ET service tool.

4.0 Benefits

- Manage and subsequently reduce tire expenses by limiting the number of tires scrapped due to heat separation.
    - One Mine Maintenance Superintendent had recently quoted:
        - *"Since implementing the TKPH system at Saraji, we reduced our tire costs by eliminating heat-related tire failures."*
- Customers can monitor tires on a proactive basis to ensure maximum life (due to the automatic monitoring of tire loading through TKPH).

- Improved machine availability

5.0 Resources Required

No additional resources are required to use the VIMS TKPH after you enable this feature.

6.0 Supporting Attachments / References

References:

- *Speed Management for CAT Mining Trucks TKPH* TEKQ0350

Attachments:

- *TKPH Monitor Set-up and Training.ppt*
  Microsoft PowerPoint file can be found in the Attachments Tab of this PDF file.

7.0 Related Best Practices

None at this time.

8.0 Acknowledgements

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